

Esercizio.

Let $\phi(x, \vec{z}, y) \in L \supseteq \epsilon_{\Delta_0}$ and let $T \supseteq ZF_L^-$ be L -theory.

- Suppose $T \vdash \forall x, \vec{z} \exists! y \phi(x, \vec{z}, y)$
- Suppose $\phi(x, \vec{z}, y) \in \Delta_1(T)$
- Let $G : Ord \times V \rightarrow V$ be the function defined by recursion using ϕ , that is:

$$G(\alpha, \vec{b}) = c \iff \phi(G \upharpoonright (\alpha \times \vec{b}), \vec{b}, c)$$

- Let θ be the graph of G .

Then $\theta(x, \vec{z}, y) \in \Delta_1(T)$.

Dimostrazione.

Basically we will write $\theta(x, \vec{z}, y)$ as:

- "there exists a function w such that $(x, \vec{z}) \in dom(w)$ and $w(x, \vec{z}) = y$ and w is the approximation of G at some ordinal β "
- "for all functions w such that $(x, \vec{z}) \in dom(w)$ and w is approximation of G at some ordinal β we have $w(x, \vec{z}) = y$ "

More formally we can write the above as:

$$(x, \vec{z}, y) \in \theta \iff \exists w \left[w \text{ function} \wedge (x, \vec{z}) \in dom(w) \wedge \right. \\ \left. \exists \beta \left(\beta \in Ord \wedge dom(w) = \beta \times \{\vec{z}\} \wedge \right. \right. \\ \left. \left. \forall \alpha \in \beta \phi(w \upharpoonright (\alpha \times \{\vec{z}\}), \vec{z}, w(\alpha \smallfrown \vec{z})) \right) \right]$$

Now observe that $\forall \alpha \in \beta \phi(w \upharpoonright (\alpha \times \{\vec{z}\}), \alpha, w(\alpha \smallfrown \vec{z})) \in \Sigma_1(T)$ since $T \vdash Collection_L$. So we have that $\theta \in \Sigma_1(T)$.

$$(x, \vec{z}, y) \in \theta \iff \forall w \left[\left(w \text{ function} \wedge (x, \vec{z}) \in dom(w) \wedge \right. \right. \\ \left. \left. \exists \beta \left(\beta \in Ord \wedge dom(w) = \beta \times \{\vec{z}\} \wedge \right. \right. \right. \\ \left. \left. \left. \forall \alpha \in \beta \phi(w \upharpoonright (\alpha \times \{\vec{z}\}), \vec{z}, w(\alpha \smallfrown \vec{z})) \right) \right) \right. \\ \left. \rightarrow y = w(x, \vec{z}) \right]$$

Now the right hand side is equivalent to:

$$\forall w \left[w \text{ not a function } \vee (x, \vec{z}) \notin \text{dom}(w) \vee y = w(x, \vec{z}) \vee \right. \\
\forall \beta \left(\beta \notin \text{Ord} \vee \text{dom}(w) \neq \beta \times \{\vec{z}\} \vee \right. \\
\left. \left. \exists \alpha \in \beta (\neg \phi(w \upharpoonright (\alpha \times \{\vec{z}\}), \vec{z}, w(\alpha, \vec{z}))) \right) \right]$$

This is $\Pi_1(T)$ since $T \vdash \text{Collection}_L$.

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